Series Solutions of ODEs

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Standard Form of the ODE

- General second order linear homogeneous ODE:

$$a_0(x)y'' + a_1(x)y' + a_2(x)y = 0.$$

- Dividing by $a_0(x)$, we write it in standard form:

$$y'' + P(x)y' + Q(x)y = 0,$$

where $P(x) = \dfrac{a_1(x)}{a_0(x)}, \quad Q(x) = \dfrac{a_2(x)}{a_0(x)}.$

- Nature of solution near $x = x_0$ depends on behaviour of $P(x)$ and $Q(x)$.
- This motivates classifying x_0 as ordinary or singular.
- A point x_0 is called analytic point of f, if it has Taylor series expansion around x_0.
- A point $x = x_0$ is called an *ordinary point* if

$$a_0(x_0) \neq 0$$



Ordinary Point: Definition

- A point $x = x_0$ is called an **ordinary point** of

$$y'' + P(x)y' + Q(x)y = 0$$

  if both $P(x)$ and $Q(x)$ are analytic at x_0.
- That is, $P(x)$ and $Q(x)$ can be expanded in a power series about x_0 with a nonzero radius of convergence.
- Equivalently, $P(x)$ and $Q(x)$ have no singularities (poles) at x_0.
- Example: In $y'' + xy' + y = 0$, every point is an ordinary point.
- At an ordinary point, we seek a solution in the form of an ordinary power series

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n.$$



Singular Point

- A point $x = x_0$ is called a **singular point** if it is not ordinary.
- That is, at least one of $P(x)$, $Q(x)$ fails to be analytic at x_0.
- Singular points are further classified as regular or irregular.
- x_0 is a **regular singular point** if $P(x)$, $Q(x)$ are not analytic at x_0.
- but $(x - x_0)P(x)$ and $(x - x_0)^2 Q(x)$ are both analytic at x_0.
- Otherwise x_0 is called an **irregular singular point**.
- Example: $x^2y'' + xy' + (x^2 - n^2)y = 0$ has a regular singular point at $x = 0$.
- Regular singular points are treated using the Frobenius method.



Existence Theorem for Series Solutions

- **Theorem:** If x_0 is an ordinary point of

$$y'' + P(x)y' + Q(x)y = 0$$

then the general solution can be written as a power series

$$y = \sum_{n=0}^{\infty} a_n (x - x_0)^n.$$

- This series converges at least for $|x - x_0| < R$, where R is the distance from x_0 to the nearest singular point.
- The two arbitrary constants a_0, a_1 correspond to two linearly independent solutions.
- For simplicity, the ordinary point $x_0 = 0$ is taken in our examples.



Method: Steps to Find Series Solution

- Step 1: Assume $y = \sum_{n=0}^{\infty} a_n x^n$ about the ordinary point $x_0 = 0$.
- Step 2: Differentiate term by term to get y' and y''.
- Step 3: Substitute y, y', y'' into the given ODE.
- Step 4: Shift indices so that all series have the same power of x.
- Step 5: Equate the coefficient of each power of x to zero.
- Step 6: Obtain a recurrence relation connecting the a_n's
- Step 7: Use the recurrence relation to express all coefficients in terms of a_0, a_1
- Step 8: Write the two linearly independent series solutions



Series Solution of a Second Order ODE

- If $y = \sum_{n=0}^{\infty} a_n x^n$, then

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}, \qquad y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

- Re-indexing y'' with $n \to n+2$:

$$y'' = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n.$$

- This shifted form is essential before comparing coefficients.
- The resulting recurrence relation typically connects a_{n+2} with a_n (or a_{n-1}).
- Solving it gives all coefficients in terms of a_0 and a_1.



Example 1: $y'' - y = 0$ (Setup)

- Solve $y'' - y = 0$ near the ordinary point $x = 0$
- Assume $y = \sum_{n=0}^{\infty} a_n x^n$
- Then $y'' = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$
- Substituting into the ODE:

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n - \sum_{n=0}^{\infty} a_n x^n = 0$$

- Equating coefficients of x^n:

$$(n+2)(n+1) a_{n+2} - a_n = 0$$

- Recurrence relation: $a_{n+2} = \dfrac{a_n}{(n+2)(n+1)}$



Example 1: $y'' - y = 0$ (Solution)

- Using $a_{n+2} = \dfrac{a_n}{(n+2)(n+1)}$ with a_0, a_1 arbitrary
- Even coefficients: $a_2 = \dfrac{a_0}{2!}$, $a_4 = \dfrac{a_0}{4!}$, $\ldots$, $a_{2k} = \dfrac{a_0}{(2k)!}$
- Odd coefficients: $a_3 = \dfrac{a_1}{3!}$, $a_5 = \dfrac{a_1}{5!}$, $\ldots$, $a_{2k+1} = \dfrac{a_1}{(2k+1)!}$
- General solution:

$$y = a_0 \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!} + a_1 \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}$$

- That is, $y = a_0 \cosh x + a_1 \sinh x$
- Converges for all x since there are no singular points.
- Note that RC of both functions is ∞. $(R = lim \frac{|a_n|}{|a_{n+1}|})$



Example 2: $y'' + y = 0$ (Setup)

- Solve $y'' + y = 0$ near the ordinary point $x = 0$.
- Assume $y = \sum_{n=0}^{\infty} a_n x^n$, so $y'' = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$.
- Substituting into the ODE:

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=0}^{\infty} a_n x^n = 0.$$

- Equating coefficients of x^n to zero:

$$(n+2)(n+1) a_{n+2} + a_n = 0.$$

- Recurrence relation: $a_{n+2} = -\dfrac{a_n}{(n+2)(n+1)}$.



Example 2: $y'' + y = 0$ (Solution)

- Even coefficients: $a_2 = -\dfrac{a_0}{2!},\ a_4 = \dfrac{a_0}{4!},\ a_{2k} = \dfrac{(-1)^k a_0}{(2k)!}$
- Odd coefficients: $a_3 = -\dfrac{a_1}{3!},\ a_5 = \dfrac{a_1}{5!},\ a_{2k+1} = \dfrac{(-1)^k a_1}{(2k+1)!}$
- General solution:

$$y = a_0 \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} + a_1 \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

- That is, $y = a_0 \cos x + a_1 \sin x$.
- Series converges for all $x \in \mathbb{R}$.

